INCH-POUND

MIL-PRF-1/1096E
6 June 2006
SUPERSEDING
MIL-E-1/1096D
24 February 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPE 5784WB

Inactive for new design
after 18 June 2001.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Pentode, subminiature, dual control.
Outline --- 3-6 (EIA).
Base --- Pinch press 7 leads in line.
Envelope --- T3.
Cathode --- Coated unipotential.
Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	a (colored dot)	g2	h	h	g3	k	g1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Ic1	Rk
Unit:	V	V dc	V dc	V dc	V dc	v	mA dc	ohms
Maximum:	6.9	165	0	155	30	200	1.0	---
Minimum:	5.7	---	-55	---	-55	---	---	---
Test conditions:	6.3	120	0	120	0	0	---	230

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Rg1	Ik	Pp	Pg2	Ic3	TE	Alt
Unit:	MegΩ	mA dc	W	W	mA dc	°C	ft
Maximum:	1.2	16.5	--- (see note 8)	--- (see note 8)	0.2	220	(see note 2)
Minimum:	---	---	---	---	---	---	---
Test conditions:	---	---	---	---	---	---	---

GENERAL:
Qualification: Not required.
Reliable tube.

AMSC N/A FSC 5960

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MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (see note 12)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 1							
1256	Electrode current (1) (anode)			0.65	Ib	3.9	7.1	mA dc
1266	Total grid current	9		0.65	Ic1	---	-0.3	µA dc
1301	Heater current			0.65	If	190	210	mA
1306	Transconductance (1)			0.65	Sm	2,650	3,950	µmhos
1336	Heater-cathode leakage			0.65	Ihk	---	5.0	µA dc
1201	Short and discontinuity detection			0.4	---	---	---	---
	Conformance inspection, part 2							
1211	Insulation of electrodes			2.5	R	100	---	MegΩ
1031	High-frequency vibration		Rp = 10,000 ohms	2.5	Ep	---	75	mV ac
1246	Audio frequency noise	3	Esig = 70 mV ac; Rg1 = 0.1 MegΩ; Ec2 = 19 V dc; Rg2 = 1,000 ohms; Rp = 0.2 MegΩ	2.5	EB	---	---	---
1256	Electrode current (2) (anode)	3	Ec3 = -10 V dc	2.5	Ib	---	200	µA dc
1256	Electrode current (3) (anode)	3	Ec3 = -6 V dc	2.5	Ib	5	---	µA dc
1256	Electrode current (4) (anode)		Ec1 = -9 V dc	2.5	Ib	---	200	µA dc
1256	Electrode current (5) (anode)		Ec1 = -5 V dc	2.5	Ib	5	---	µA dc
1256	Electrode current (screen grid)			2.5	Ic2	2.8	5.4	mA dc
1266	Grid emission	4	Ef = 7.5 V; Ec1 = -10 V dc	6.5	Ic1	---	-0.5	µA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	$\frac{\Delta Sm}{Ef}$	---	10	%

See notes at end of table.

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MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (see note 12)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 2 - Continued							
1306	Transconductance (3)	3	Ec3 = -1.0 V dc	6.5	Sg3-a	400	1,100	μmhos
1306	Transconductance (4)	3	Ec3 = 22 V dc	6.5	Sg3-a	---	25	μmhos
1331	Direct-interelectrode capacitance		0.405 inch diameter shield	6.5	Cg1p Cin Cout	--- 3.5 2.8	0.030 5.5 4.4	pF pF pF
1256	Electrode current (1) (anode) (tube operation time)		t = 20 seconds	4.0	Ib	90	110	%
1031	Sweep-frequency vibration	5	Rp = 10,000 ohms; F = 30 to 1,000 Hz; 15 G; t = 3 minutes; position X and Y only	4.0	ep	---	300	mv p to p
1116	Lead fatigue			2.5	---	---	---	---
2126	Envelope strain			2.5	---	---	---	---
1041	Shock	6	450 G; Ehk = 100 Vdc	---	---	---	---	---
1105	Permanence of marking			---	---	---	---	---
1119	Glass-to-lead seal strain			6.5	---	---	---	---
1031	Vibration-fatigue (1)			6.5	---	---	---	---
1031	Vibration fatigue (2)		10G; t = 6 hours	6.5	---	---	---	---
---	Shock and vibration-fatigue (1) and (2) - test end points:							
1031	High-frequency vibration			---	Ep	---	125	mV ac
1336	Heater-cathode leakage			---	Ihk	---	20	μA dc
1306	Change in trans-conductance (1) of individual tubes			---	$\frac{\Delta Sm}{t}$	---	15	%
1266	Total grid current			---	Ic1	---	-10	μA dc

See notes at end of table.

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MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5; Ehk = 140 V ac; Eb = Ec1 = Ec2 = Ec3 = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1516	Stability life		Ehk = 200 V dc; Rg1 = 1.0 MegΩ; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			$\frac{\Delta Sm}{t}$	---	10	%
1501	Intermittent life	7, 11	Ehk = 200 V dc; Rg1 = 1.0 MegΩ; TE = 220°C (min)	---	---	---	---
---	Intermittent life-test end points (500 hours):	8					
1266	Total grid current			Ic1	---	-0.6	µA dc
1301	Heater current			If	188	212	mA
1306	Change in transconductance (1) of individual tubes			$\frac{\Delta Sm}{Ef}$	---	20	%
1306	Transconductance (2)			$\frac{\Delta Sm}{Ef}$	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	µA dc
1306	Transconductance (1), average change			Avg $\frac{\Delta Sm}{t}$	---	15	%
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic1	---	-1.0	µA dc
1301	Heater current			If	188	212	mA
1306	Change in transconductance (1) of individual tubes			$\frac{\Delta Sm}{t}$	---	25	%
1336	Heater-cathode leakage			Ihk	---	15	µA dc

See notes at top of next page.

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NOTES:

1. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. The reference point for Ec3 shall be the negative side of the cathode resistor.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test shall not be permitted. Without removing the tube from the preheat socket, the preheat conditions shall be applied, except Ec1 = -10 V, and the grid current reading made within 3 seconds after the end of the preheat period. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Eb	Ec1	Ec2	Ec3	Rk	Rg1
V	V dc	V dc	V dc	V dc	ohms	MegΩ
7.5	120	0	120	0	230	1.0

5. The impedance of the anode voltage supply shall not exceed that of a 40-µF capacitor at 10 Hz.
6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
8. The design application maximum rating of the tube shall not exceed 0.79 watts per anode power dissipation and shall not exceed 0.6 watts for grid No. 2 power dissipation.
9. This test shall be performed at the conclusion of the holding period.
10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.
11. The life-test sample shall consist of 10 tubes per lot and no tube failure shall be permitted. In the event of rejection of the first sample, due to failure of 1 tube, a second sample of 20 tubes shall be selected from the lot. Acceptance shall then be based on the second sample, which shall be tested with no failures.
12. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Referenced documents: In addition to MIL-PRF-1, this document references the following: MIL-STD-1311

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2006-009)

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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